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No. 117.—VOL. III.

SATURDAY, MARCH 27, 1886.

PRICE 1½d.

GRETNA GREEN AND ITS MARRIAGES.

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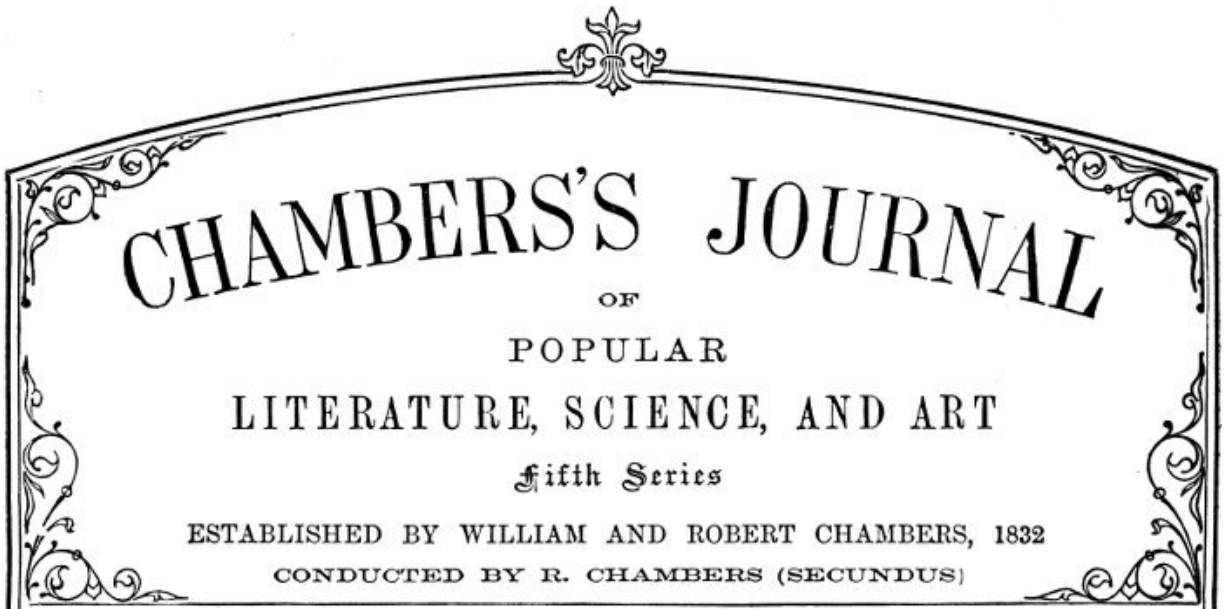
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these lack nought of the legal validity and obligation of those contracted with pealing organ and the most ceremoniously conducted ecclesiastical display. The Act of 1856 only makes Scotch marriages illegal in the case of one or other of the parties not having resided for three weeks in the kingdom of Scotland, thereby putting a stop to many runaway marriages, especially among servants, who came across in numbers from Carlisle at the season of annual hirings.

Not very long since, a faithless swain, weary prematurely of vows exchanged at Gretna Green, and doubting somewhat, it may be, of the holiness of the estate inaugurated by rites so maimed, betook himself, in the company of another and, to him, doubtless fairer bride, to a Roman Catholic priest in a southern Scottish burgh, who all unwittingly solemnised a marriage between them, destined to work no small evil to the fickle bridegroom; for mark how well the sequel hangs together. The deceiver, a sadder and perchance wiser man, torn from the arms of his too credulous bride, a Niobe all tears, was hauled before the outraged majesty of law, and compelled to undergo the penalties, not trivial, awarded to crimes of perjury and bigamy.

Whatever peculiar popularity as a marriage-resort may have been enjoyed by Gretna Green is doubtless due to the convenience and accessibility of its situation on the Great North Road; for here is no instance of especial virtue residing in local fountains, but merely of such virtue—if, indeed, one may so use the term—as is participated in by every other spot of ground within the whole realm of Scotland; nor, indeed, as a matter of fact, were Coldstream and Lamberton near Berwick without some measure of peculiar advantage, which they offered to those impatient ones who, from the more eastern counties, were minded to avail themselves of the proximity of the Scottish Border.

The origin of these marriages has been sought by some in the wild habits of times far distant, when lack of clergy in the district was to some extent supplied by the ministrations of friars from the adjacent abbeys of Melrose and Jedburgh, who in the course of their perambulations performed the rites of baptism and marriage. The Borderer, nowise forgetful, ere setting forth on expeditions of rapine and plunder, to tell his beads right zealously, was yet grossly ignorant about many things; nor had he access to any other source of enlightenment than the ‘Book-a-bosoms,’ as the mass-book was

called, from the habit of the wandering ecclesiastics carrying it in their bosoms. Thus it was that stout William of Deloraine seemed, to the astonished eyes of the Goblin Page, so strangely to resemble one of these friars, when

As the corselet off he took,
The Dwarf espied the mighty Book!
Much he marvelled, a knight of pride
Like a book-bosomed priest should ride.

But it may have been that this custom originated at Gretna Green about 1738, on the suppression of the infamous Fleet marriages, though, without doubt, irregular marriage was far from unknown long prior to this time in the Border parishes. At all events, acting on his knowledge that Scotch marriages, where parties accepted each other as man and wife before witnesses, were legal, one Scott opened a place at the Rigg, in the parish of Gretna, and there marriages were celebrated between runaway couples about the year 1753. Scott was succeeded by an old soldier named Gordon, who was wont to officiate in uniform, wearing a huge cocked-hat, and girt about the waist with a ponderous sword.

In 1842 were published by Robert Elliott the *Gretna Green Memoirs*, wherein we are told how Elliott—a retired stagecoach driver—became acquainted with Joseph Paisley, successor of the veteran Gordon in 1810. Paisley, who had been a tobacconist, fisherman, nay, even, it is more than hinted, a smuggler, became known as ‘the blacksmith,’ from the speed with which he riveted the bonds of runaway couples. Elliott, who married Paisley’s daughter, and eventually succeeded him in his office, continued sole and only ‘parson’ of Gretna Green for twenty-nine years, during which period he is said to have united more than three thousand couples of all ranks and grades in society, the greatest number in any one year (1825) having been one hundred and ninety-eight, and the average from 1829 to 1835 inclusive upwards of one hundred and sixty each year. Although tradition says that Lord Erskine paid as much as eighty guineas on the occasion of his marriage, the average fee at Gretna Green is estimated at fifteen guineas; whence we may at anyrate infer how much more highly paid was the Border ‘parson’ than the majority of the more regular clergy on either side of the Sark.

In a will-case tried some years ago at Liverpool, the plaintiff, Robert Ker, had been married on two occasions at Gretna Green—in 1850, and again in 1853—the first marriage having been solemnised in a beerhouse at Springfield, near Gretna; and the second in an alehouse kept by William Blythe, when Thomas Blythe, in presence of his wife, performed the ceremony, which was thus described: ‘I went in and had some conversation, and asked him [Thomas Blythe] to do this little job. He said he would, and asked me if I was willing to take this lady as my wife, and I said yes. Then he asked her if she was willing to take me for her husband, and she said she was; and I got hold of her hand and put the ring on, and we were declared man and wife; and that was how we were married.’ At this trial, a book containing a register of marriages performed by the Blythes was produced in evidence.

Thomas Blythe was himself examined in the Probate Court at Westminster, and stated that in the May of 1853 he was living at Springfield, Gretna Green, and was in the agricultural line, though he did a small stroke of business in the ‘joining line’ as well. Replying to counsel as to how he performed the ceremony, he gave the following account of the marriage service as by him conducted: ‘I first asked if they were single. They said they were. I then asked the man: “Do you take this woman for your wife?” He said, “Yes.” I then asked the woman: “Do you take this man for your lawful husband?” She said, “Yes.” I then said: “Put on the ring.” The ring was put on. I then said: “The thing is done; the marriage is complete.”’ A certificate of marriage was written out and given to the woman.

We doubt not, however, that many of our readers may learn with surprise that, even now, marriage—provided that one or other of the parties have resided three weeks in Scotland—may be thus speedily and effectually performed at the erstwhile notorious little village of Gretna Green, as well as elsewhere north of the Border.

IN ALL SHADES.

CHAPTER XV.

A FORTNIGHT after Nora's arrival in Trinidad, Mr Tom Dupuy, neatly dressed in all his best, called over one evening at Orange Grove for the express purpose of speaking seriously with his pretty cousin. Mr Tom had been across to see her more than once already, to be sure, and had condescended to observe to many of his men acquaintances, on his return from his call, that Uncle Theodore's girl, just come out from England, was really in her own way a most elegant and attractive creature. In Mr Tom's opinion, she would sit splendidly at the head of the table at Pimento Valley. 'A man in my position in life wants a handsome woman, you know,' he said, 'to do the honours, and keep up the dignity of the family, and look after the women-servants, and all that sort of thing; so Uncle Theodore and I have arranged beforehand that it would be a very convenient plan if Nora and I were just to go and make a match of it.'

With the object of definitely broaching this preconcerted harmony to his unconscious cousin, Mr Tom had decked himself in his very smartest coat and trousers, stuck a *gloire de Dijon* rose in his top button-hole, mounted his celebrated gray Mexican pony 'Sambo Gal,' and ridden across to Orange Grove in the cool of the evening.

Nora was sitting by herself with her cup of tea in the little boudoir that opened out on to the terrace garden, with its big bamboos and yuccas and dracæna trees, when Mr Tom Dupuy was announced by Rosina as waiting to see her.

'Show him in, Rosina,' Nora said with a smile; 'and ask Aunt Clemmy to send me up another teacup.—Good-evening, Tom. I'm afraid you'll find it a little dull here, as it happens, this evening, for papa's gone down to Port-of-Spain on business; and so you'll have nobody to talk with you to-night about the prospects of the year's sugar-crop.'

Tom Dupuy seated himself on the ottoman beside her with cousinly liberty. 'Oh, it don't matter a bit, Nora,' he answered with his own peculiar gallantry. 'I don't mind. In fact, I came over on purpose this evening, knowing Uncle Theodore was out, because I'd got something very particular I wanted to talk over with you in private.'

'In-deed,' Nora answered emphatically. 'I'm surprised to hear it. I assure you, Tom, I'm absolutely ignorant on the subject of cane-culture.'

'Girls brought up in England mostly are,' Tom Dupuy replied with the air of a man who generously makes a great concession. 'They don't appear to feel much interest in sugar, like other people. I suppose in England there's nothing much grown except corn and cattle.—But that wasn't what I came over to talk about to-night, Nora. I've got something on my mind that Uncle Theodore and I have been thinking over, and I want to make a proposition to you about it.'

'Well, Tom?'

'Well, Nora, you see, it's like this. As you know, Orange Grove is Uncle Theodore's to leave; and after his time, he'll leave it to you, of course; but Pimento Valley's entailed on me; and that being so, Uncle Theodore lets me have it on lease during his lifetime, so that, of course, whatever I spend upon it in the way of permanent improvements is really spent in bettering what's practically as good as my own property.'

'I understand. Quite so.—Have a cup of tea?'

'Thank you.—Well, Pimento Valley, you know, is one of the very best sugar-producing estates in the whole island. I've introduced the patent Browning regulators for the centrifugal process; and I've imported some of these new Indian mongooses that everybody's talking about, to kill off the cane-rats; and I've got some splendid stock rattoons over from Mauritius; and altogether, a finer or more creditable irrigated estate I don't think you'll find—though it's me that says it—in the island of Trinidad. Why, Nora, at our last boiling, I assure you the greater part of the liquor turned out to be seventeen over proof; while the molasses stood at twenty-nine specific gravity; giving a yield, you know, of something like one hogshead decimal four on the average to the acre of canes under cultivation.'

Nora held up her fan carelessly to smother a yawn. 'I daresay it did, Tom,' she answered with obvious unconcern; 'but, you know, I told you I didn't understand anything on earth about sugar; and you said it wasn't about that that you wanted to talk to me in private this evening.'

'Yes, yes, Nora; you're quite right; it isn't. It's about a far deeper and more interesting subject than sugar that I'm going to speak to you.' (Nora mentally guessed it must be rum.) 'I only mentioned these facts, you see, just to show you the sort of yield we're making now at Pimento Valley. Last year, we did five hundred hogsheads, and two hundred and eighty-four puncheons. A man who does a return like that, of course, must naturally be making a very tidy round little income.'

'I'm awfully glad to hear it, I'm sure, for your sake,' Nora answered unconcernedly.

'I thought you would be, Nora; I was sure you would be. Naturally, it's a matter that touches us both very closely. You see, as you're to inherit Orange Grove, and as I'm to inherit Pimento Valley, Uncle Theodore and I think it would be a great pity that the two old estates—the estates bound up so intimately with the name and fame of the fighting Dupuys—should ever be divided or go out of the family. So we've agreed together, Uncle Theodore and I, that I should—well, that I should endeavour to unite them by mutual arrangement.'

'I don't exactly understand,' Nora said, as yet quite unsuspecting of his real meaning.

'Why, you know, Nora, a man can't live upon sugar and rum alone.'

'Certainly not,' Nora interrupted; 'even if he's a confirmed drunkard, it would be quite impossible. He must have something solid occasionally to eat as well.'

'Ah, yes,' Tom said, in a sentimental tone, endeavouring to rise as far as he was able to the height of the occasion. 'And he must have something more than that too, Nora: he must have sympathy; he must have affection: he must have a companion in life; he must have somebody, you know, to sit at the head of his table, and to—to—to'——

'To pour out tea for him,' Nora suggested blandly, filling his cup a second time.

Tom reddened a little. It wasn't exactly the idea he wanted, and he began to have a faint undercurrent of suspicion that Nora was quietly laughing at him in her sleeve. 'Ah, well, to pour out tea for him,' he went on, somewhat suspiciously; 'and to share his joys and sorrows, and his hopes and aspirations'——

'About the sugar-crop?' Nora put in once more, with provoking calmness.

'Well, Nora, you may smile if you like,' Tom said warmly; 'but this is a very serious subject, I can tell you, for both of us. What I mean to say is that Uncle Theodore and I have settled it would be a very good thing indeed if we two were to get up a match between us.'

'A match between you,' Nora echoed in a puzzled manner—'a match between papa and you, Tom! What at? Billiards? Cricket? Long jumping?'

Tom fairly lost his temper. 'Nonsense, Nora,' he said testily. 'You know as well what I mean as I do. Not a match between Uncle Theodore and me, but a match between you and me—the heir and heiress of Orange Grove and Pimento Valley.'

Nora stared at him with irrepressible laughter twinkling suddenly out of all the corners of her merry little mouth and puckered eyelids. 'Between you and me, Tom,' she repeated incredulously—'between you and me, did you say? Between you and me now? Why, Tom, do you really mean this for a sort of an offhand casual proposal?'

'Oh, you may laugh if you like,' Tom Dupuy replied evasively, at once assuming the defensive, as boors always do by instinct under similar circumstances. 'I know the ways of you girls that have been brought up at highfalutin' schools over in England. You think West Indian gentlemen aren't good enough for you, and you go running after cavalry-officer fellows, or else after some confounded upstart woolly-headed mulatto or other, who come out from England. I know the ways of you. But you may laugh as you like. I see you don't mean to listen to me now; but you'll have to listen to me in the end; for Uncle Theodore and I have made up our minds about it, and what a Dupuy makes up his mind about, he generally sticks to, and there's no turning him. So in the end, I know, Nora, you'll have to marry me.'

‘You seem to forget,’ Nora said haughtily, ‘that I too am a Dupuy, as much as you are.’

‘Ah, but you’re only a woman, and that’s very different. I don’t mind a bit about your answering me *no* to-day. It seems I’ve tapped the puncheon a bit too early; that’s all: leave the liquor alone, and it’ll mature of itself in time in its own cellar. Sooner or later, Nora, you see if you don’t marry me.’

‘But, Tom,’ Nora cried, abashed into seriousness for a moment by his sudden outburst of native vulgarity, ‘this is really so unexpected and so ridiculous. We’re cousins, you know; I’ve never thought of you at all in any way except as a cousin. I didn’t mean to be rude to you; but your proposal and your way of putting it took me really so much by surprise.’

‘Oh, if that’s all you mean,’ Tom Dupuy answered, somewhat mollified, ‘I don’t mind your laughing, no, not tuppence. All I mind is your saying *no* so straight outright to me. If you want time to consider’——

‘Never!’ Nora interrupted quickly in a sharp voice of unswerving firmness.

‘Never, Nora? Never? Why never?’

‘Because, Tom, I don’t care for you; I can’t care for you; and I never will care for you. Is that plain enough?’

Tom stroked his chin and looked at her dubiously, as a man looks at an impatient horse of doubtful temper. ‘Well,’ he said, ‘Nora, you’re a fine one, you are—a very fine one. I know what this means. I’ve seen it before lots of times. You want to marry some woolly-headed brown man. I heard you were awfully thick with some of those people on board the *Severn*. That’s what always comes of sending West Indian girls to be educated in England. You’ll have to marry me in the end, though, all the same, because of the property. But you just mark my words: if you don’t marry me, as sure as fate, you’ll finish with marrying a woolly-headed mulatto!’

Nora rose to her full height with offended dignity. ‘Tom Dupuy,’ she said angrily, ‘you insult me! Leave the house, sir, this minute, or I shall retire to my room. Get back to your sugar-canes and your centrifugals until you’ve learned better manners.’

‘Upon my word,’ Tom said aloud, as if to himself, rising to go, and flicking his boot carelessly with his riding-whip, ‘I admire her all the more when

she's in a temper. She's one of your high-steppers, she is. She's an uncommon fine girl, too—hanged if she isn't—and, sooner or later, she'll have to marry me.'

Nora swept out of the boudoir without another word, and walked with a stately tread into her own room. But before she got there, the ludicrous side of the thing had once more overcome her, and she flung herself on a couch in uncontrollable fits of childish laughter. 'Oh, Aunt Clemmy,' she cried, 'bring me my tea in here, will you? I really think I shall die of laughing at Mr Tom there!'

CHAPTER XVI.

For a few days, the Hawthorns had plenty of callers—but all gentlemen. Marian did not go down to receive them. Edward saw them by himself in the drawing-room, accepting their excuses with polite incredulity, and dismissing them as soon as possible by a resolutely quiet and taciturn demeanour. Such a singularly silent man as the new judge, everybody said, had never before been known in the district of Westmoreland.

One afternoon, however, when the two Hawthorns were sitting out under the spreading mango-tree in the back-garden, forgetting their doubts and hesitations in a quiet chat, Thomas came out to inform them duly that two gentlemen and a lady were waiting to see them in the big bare drawing-room. Marian sighed a sigh of profound relief. 'A lady at last,' she said hopefully. 'Perhaps, Edward, they've begun to find out, after all, that they've made some mistake or other. Can—can any wicked person, I wonder, have been spreading around some horrid report about me, that's now discovered to be a mere falsehood?'

'It's incomprehensible,' Edward answered moodily. 'The more I puzzle over it, the less I understand it. But as a lady has called at last, of course, darling, you'd better come in at once and see her.'

They walked together, full of curiosity, into the drawing-room. The two gentlemen rose simultaneously as they entered. To Marian's surprise, it was Dr Whitaker and his father; and with them had come—a brown lady.

Marian was unaffectedly glad to see their late travelling companion; but it was certainly a shock to her, unprejudiced as she was, that the very first and

only woman who had called upon her in Trinidad should be a mulatto. However, she tried to bear her disappointment bravely, and sat down to do the honours as well as she was able to her unexpected visitors.

‘My daughtah!’ the elder brown man said ostentatiously, with an expansive wave of his greasy left hand towards the mulatto lady—‘Miss Euphemia Fowell-Buxton Duchess-of-Sutherland Whitaker.’

Marian acknowledged the introduction with a slight bow, and bit her lip. She stole a look at Dr Whitaker, and saw at once upon his face an unwonted expression of profound dejection and disappointment.

‘An’ how do you like Trinidad, Mrs Hawthorn?’ Miss Euphemia asked with a society simper; while Edward began engaging in conversation with the two men. ‘You find de excessiveness of de temperature prejudicial to salubrity, after de delicious equability of de English climate?’

‘Well,’ Marian assented smiling, ‘I certainly do find it very hot.’

‘Oh, exceedingly,’ Miss Euphemia replied, as she mopped her forehead violently with a highly scented lace-edged cambric pocket-handkerchief. ‘De heat is most oppressive, most unendurable. I could wring out me handkerchief, I assure you, Mrs Hawthorn, wit de extraordinary profusion of me perspiration.’

‘But this is summer, you must remember,’ Dr Whitaker put in nervously, endeavouring in vain to distract attention for the moment from Miss Euphemia’s conversational peculiarities. ‘In winter, you know, we shall have quite delightful English weather on the hills—quite delightful English weather.’

‘Ah, yes,’ the father went on with a broad smile. ‘In winter, Mrs Hawthorn, ma’am, you will be glad to drink a glass of rum-and-milk sometimes, I tell you, to warm de blood on dese chilly hilltops.’

The talk went on for a while about such ordinary casual topics; and then at last Miss Euphemia happened to remark confidentially to Marian, that that very day her cousin, Mr Septimius Whitaker, had been married at eleven o’clock down at the cathedral.

‘Indeed,’ Marian said, with some polite show of interest. ‘And did you go to the wedding, Miss Whitaker?’

Miss Euphemia drew herself up with great dignity. She was a good-looking, buxom, round-faced, very negro-featured girl, about as dark in complexion as her brother the doctor, but much more decidedly thick-lipped and flat-nosed. 'O no,' she said, with every sign of offended prejudice. 'We didn't at all approve of de match me cousin Septimius was unhappily makin'. De lady, I regret to say, was a Sambo.'

'A what?' Marian inquired curiously.

'A Sambo, a Sambo gal,' Miss Euphemia replied in a shrill crescendo.

'Oh, indeed,' Marian assented in a tone which clearly showed she hadn't the faintest idea of Miss Euphemia's meaning.

'A Sambo,' Mr Whitaker the elder said, smiling, and coming to her rescue—'a Sambo, Mrs Hawthorn, is one of de inferior degrees in de classified scale and hierarchy of colour. De offspring of an African and a white man is a mulatto—dat, madam, is my complexion. De offspring of a mulatto and a white man is a quadroon—dat is de grade immediately superior. But de offspring of a mulatto and a negress is a Sambo—dat is de class just beneat us. De cause of complaint alleged by de family against our nephew Septimius is dis—dat bein' himself a mulatto—de very fust remove from de pure-blooded white man—he has chosen to ally himself in marriage wit a Sambo gal—de second and inferior remove in de same progression. De family feels dat in dis course Septimius has toroughly and irremediably disgraced himself.'

'And for dat reason,' added Miss Euphemia with stately coldness, 'none of de ladies in de brown society of Trinidad have been present at dis morning's ceremony. De gentlemen went, but de ladies didn't.'

'It seems to me,' Dr Whitaker said, in a pained and humiliated tone, 'that we oughtn't to be making these absurd distinctions of minute hue between ourselves, but ought rather to be trying our best to break down the whole barrier of time-honoured prejudice by which the coloured race, as a race, is so surrounded.—Don't you agree with me, Mr Hawthorn?'

'Pho!' Miss Euphemia exclaimed, with evident disgust. 'Just listen to Wilberforce! He has no proper pride in his family or in his colour. He would go and shake hands wit any vulgar, dirty, nigger woman, I believe, as black

as de poker; his ideas are so common!—Wilberforce, I declare, I's quite ashamed of you!'

Dr Whitaker played nervously with the knob of his walking-stick. 'I feel sure, Euphemia,' he said at last, 'these petty discriminations between shade and shade are the true disgrace and ruin of our brown people. In despising one another, or boasting over one another, for our extra fraction or so of white blood, we are implicitly admitting in principle the claim of white people to look down upon all of us impartially as inferior creatures.—Don't you think so, Mr Hawthorn?'

'I quite agree with you,' Edward answered warmly. 'The principle's obvious.'

Dr Whitaker looked pleased and flattered. Edward stole a glance at Marian, and neither could resist a faint smile at Miss Euphemia's prejudices of colour, in spite of their pressing doubts and preoccupations. And yet, they didn't even then begin to perceive the true meaning of the situation. They had not long to wait, however, for before the Whitakers rose to take their departure, Thomas came in with a couple of cards to announce Mr Theodore Dupuy, and his nephew, Mr Tom Dupuy of Pimento Valley.

The Whitakers went off shortly, Miss Euphemia especially in very high spirits, because Mrs Hawthorn had shaken hands in the most cordial manner with her, before the face of the two white men. Edward and Marian would fain have refused to see the Dupuys, as they hadn't thought fit to bring even Nora with them; and at that last mysterious insult—a dagger to her heart—the tears came up irresistibly to poor wearied Marian's swimming eyelids. But Thomas had brought the visitors in before the Whitakers rose to go, and so there was nothing left but to get through the interview somehow, with what grace they could manage to muster.

'We had hoped to see Nora long before this,' Edward Hawthorn said pointedly to Mr Dupuy—after a few preliminary polite inanities—half hoping thus to bring things at last to a positive crisis. 'My wife and she were school-girls together, you know, and we saw so much of one another on the way out. We have been quite looking forward to her paying us a visit.'

Mr Dupuy drew himself up very stiffly, and answered in a tone of the chilliest order: 'I don't know to whom you can be alluding, sir, when you speak of "Nora;" but if you refer to my daughter, Miss Dupuy, I regret to say she is suffering just at present from—ur—a severe indisposition, which unfortunately prevents her from paying a call on Mrs Hawthorn.'

Edward coughed an angry little cough, which Marian saw at once meant a fixed determination to pursue the matter to the bitter end. 'Miss Dupuy herself requested me to call her Nora,' he said, 'on our journey over, during which we naturally became very intimate, as she was put in charge of my wife at Southampton, by her aunt in England. If she had not done so, I should never have dreamt of addressing her, or speaking of her, by her Christian name. As she did do so, however, I shall take the liberty of continuing to call her by that name, until I receive a request to desist from her own lips. We have long been expecting a call, I repeat, Mr Dupuy, from your daughter Nora.'

'Sir!' Mr Dupuy exclaimed angrily; the blood of the fighting Dupuys was boiling up now savagely within him.

'We have been expecting her,' Edward Hawthorn repeated firmly; 'and I insist upon knowing the reason why you have not brought her with you.'

'I have already said, sir,' Mr Dupuy answered, rising and growing purple in the face, 'that my daughter is suffering from a severe indisposition.'

'And I refuse,' Edward replied, in his sternest tone, rising also, 'to accept that flimsy excuse—in short, to call it by its proper name, that transparent falsehood. If you do not tell me the true reason at once, much as I respect and like Miss Dupuy, I shall have to ask you, sir, to leave my house immediately.'

A light seemed to burst suddenly upon the passionate planter, which altered his face curiously, by gradual changes, from livid blue to bright scarlet. The corners of his mouth began to go up sideways in a solemnly ludicrous fashion: the crow's-feet about his eyes first relaxed and then tightened deeply; his whole big body seemed to be inwardly shaken by a kind of suppressed impalpable laughter. 'Why, Tom,' he exclaimed, turning with a curious half-comical look to his wondering nephew, 'do you know—upon

my word—I really believe—no, it can't be possible—but I really believe—they don't even now know anything at all about it.'

'Explain yourself,' Edward said sternly, placing himself between Mr Dupuy and the door, as if on purpose to bar the passage outward.

'If you really don't know about it,' Mr Dupuy said slowly, with an unusual burst of generosity for him, 'why, then, I admit, the insult to Miss Dupuy is—is—is less deliberately intentional than I at first sight imagined.—But no, no: you *must* know all about it already. You can't still remain in ignorance. It's impossible, quite impossible.'

'Explain,' Edward reiterated inexorably.

'You compel me?'

'I compel you.'

'You'd better not; you won't like it.'

'I insist upon it.'

'Well, really, since you make a point of it—but there, you've been brought up like a gentleman, Mr Hawthorn, and you've married a wife who, as I learn from my daughter, is well connected, and has been brought up like a lady; and I don't want to hurt your feelings needlessly. I can understand that under such circumstances'——

'Explain. Say what you have to say; I can endure it.'

'Tom!' Mr Dupuy murmured imploringly, turning to his nephew. After all, the elder man was something of a gentleman; he shrank from speaking out that horrid secret.

'Well, you see, Mr Hawthorn,' Tom Dupuy went on, taking up the parable with a sardonic smile—for he had no such scruples—'my uncle naturally felt that with a man of *your colour*'—— He paused significantly.

Edward Hawthorn's colour at that particular moment was vivid crimson. The next instant it was marble white. 'A man of my colour!' he exclaimed, drawing back in astonishment, not unmingled with horror, and flinging up his arms wildly—'a man of my colour! For heaven's sake, sir, what, in the name of goodness, do you mean by a man of my colour?'

‘Why, of course,’ Tom Dupuy replied maliciously and coolly, ‘seeing that you’re a brown man yourself, and that your father and mother were brown people before you, naturally, my uncle’——

Marian burst forth into a little cry of intense excitement. It wasn’t horror; it wasn’t anger; it wasn’t disappointment: it was simply relief from the long agony of that endless, horrible suspense.

‘We can bear it all, Edward,’ she cried aloud cheerfully, almost joyously —‘we can bear it all! My darling, my darling, it is nothing, nothing, nothing!’

And regardless of the two men, who stood there still, cynical and silent, watching the effect of their unexpected thunderbolt, the poor young wife flung her arms wildly around her newly wedded husband, and smothered him in a perfect torrent of passionate kisses.

But as for Edward, he stood there still, as white, as cold, and as motionless as a statue.

(To be continued.)

CANAL NAVIGATIONS.

UNTIL the middle of the last century, our forefathers thought far more of foreign enterprise than of the internal communications of their own island. An Englishman of the time of Elizabeth might be acquainted with all the intricacies of the Arctic Ocean or of the West Indies; but it by no means followed that he was able to sketch a map of his own country. The sea was the great highway of trade and fame, and the commercial towns were all seaports.

Previous to the accession of George III., the communications throughout England were of the most wretched kind, the great highways being simply the worn-out tracks of the old Roman roads. The manufactures of our country, struggling into notice, were greatly hampered by this lack of communication, few facilities for carriage existing, and distant markets being beyond reach. The little carrying-trade was necessarily of the slowest and most expensive kind, and goods were conveyed to the nearest port or navigable river, generally by long strings of packhorses, less frequently by the slow clumsy stage-wagon. Packhorses conveyed from the Severn the clay used in the Potteries, bringing back in return coarse earthenware for export. The cloth-manufacturer of Yorkshire saddled his horse with his wares and travelled from fair to fair as his own salesman; and the little cotton used in the Manchester looms was transported from Liverpool in the same primitive fashion.

This was the state of the communications in England in 1757, when the Duke of Bridgewater, having been crossed in love by one of the beautiful Miss Gunnings, turned his attention to the more prosaic employment of canal construction. His idea was to construct a waterway, or 'navigation,' from his coal-pits to Manchester, a distance of ten miles. Short as this distance appears in our time, it offered so great a barrier in those days, that the supply of fuel was always limited and uncertain. The duke, who was desirous of engaging an engineer to put his idea into practical form, was advised to employ the famous millwright Brindley, who had already made himself a name in the district for his clever contrivances in the pottery-

works and the silk-factories. Like many others who have risen to fame, Brindley was a self-made man. To his natural-born genius, there were united two characteristics which are necessary to all such pioneers—great perseverance, and a confidence in his own judgment which overbore all the adverse criticism of the multitude. His diary, which is extant, shows his school education to have been of the scantiest; the words, spelt in the broad Staffordshire dialect, and the painfully crabbed writing, excite alternately our amusement and our respect; whilst it shows throughout the dogged determination of the individual to overcome difficulty.

Brindley was no sooner installed as engineer of the works than he completely altered the duke's plan. To construct the proposed canal—or 'novogation,' as Brindley has it—it was necessary to cross the river Irwell, and it was here that he first showed his marvellous courage and skill. The duke's plan had been to drop the canal by a series of locks to the level of the river, and to raise it again on the farther side by the same means. But Brindley, who foresaw that locks would always prove a great hindrance to traffic, decided that the canal should not change its level, but should cross the river on a stone aqueduct. Nothing of the kind had ever before been attempted in this country, and, to ordinary minds, the idea of boats, laden with coals, sailing, as it were in mid-air, seemed preposterous. It must be allowed, to the everlasting credit of the duke, that, although somewhat uncertain in his own mind as to the result of the scheme, he nevertheless allowed Brindley to proceed. In spite of general ridicule, the works were commenced, the aqueduct was built; and derision was turned into amazement when the canal-boats passed over and the structure showed no sign of collapse. The packhorses were dispensed with, and the price of coal in Manchester fell to one-half. The success, both to the projector and the community, was so complete, that the duke at once sought further powers to extend the canal westward, and thus to open communication with the port of Liverpool. After much opposition from landowners and others, Brindley commenced this extension; but although no great engineering difficulties were encountered, the expenditure for some years had been so heavy that the want of money threatened to offer a serious obstacle to the completion of the scheme. The duke's credit became so low that the greatest task of the week was the collecting of a sufficient amount to pay the wages of the labourers on the works; and it was only by much scheming and economy that the works were at length completed.

Meanwhile, the Staffordshire Potteries had begun to clamour for a waterway, and Brindley had undertaken the survey of a canal which was to connect them with the Trent and Mersey. Wedgwood, the great potter, gave all his influence to a scheme for uniting his factories with the sea, and even removed his works to a site on the proposed canal, known henceforth by the ancient name of Etruria. The great undertaking in the construction of this canal was the tunnel, a mile and a half in length, under that part of the Pennine chain which separates Staffordshire from Cheshire. This tunnel was to constitute the highest point or 'summit-level' of the canal; and the supply of water was to be obtained from a system of reservoirs situated at a still higher elevation and fed by the surrounding hills. But tunnelling was a new experiment in engineering; many unforeseen difficulties arose to hinder the work, and it was only after eleven years of heavy anxiety and stubborn perseverance that this last link in the communication was completed. The carriage of a ton of goods from Liverpool to Etruria, which had cost under the old system fifty shillings, was reduced to one-fourth. This tunnel, the pioneer of many miles of tunnelling since constructed, still exists. It is simply a long culvert, just large enough to allow of the passage of a single barge. There is no accommodation for hauling the traffic through, and the barges are consequently propelled from end to end by the exertions of the boatmen alone. Fifty years after its construction, the traffic on the canal had increased to such an extent that the mouths of the tunnel were perpetually blocked by a crowd of boats waiting to pass through, and the fights and quarrels among the boatmen for first place were a disgrace to the Canal Company. After much pressure, the authorities called in the Scotch engineer Telford, and to him was intrusted the construction of a second tunnel. The want of suitable machinery, of skilled labour and of money, were obstacles comparatively unknown to Telford, and the new tunnel, large enough to allow of a towing-path, was constructed in three years. The two works, side by side, represent fifty years' progress in the science of engineering.

But to return to Brindley and his triumphs. In North Warwickshire, a colony of iron-workers had sprung up in the midst of a plain, worn into narrow 'hollow-ways' by the tread of the ubiquitous packhorse. The few letters sent to this large village of blacksmiths were addressed 'Birmingham, near Coleshill,' this latter place being the nearest point on the high road. Through this district, Brindley succeeded in cutting a canal from the Trent to the

Severn; and thus Birmingham, the Potteries, and Manchester were each connected with the Irish and North Seas.

Brindley's last great work was the projection of a canal from Leeds to Liverpool; but owing partly to the difficulties of the country passed through, and partly to the scarcity of labourers through the continental wars, the canal was not completed throughout until 1816, long after Brindley's death. The summit of this canal is in the wild and stony district of Pendle Forest, where are situated the great reservoirs—one being over a hundred acres in extent—which feed the higher levels of the canal with water. These reservoirs are maintained in repair and efficiency at the present day by the owners of the numerous stone quarries of the district, to whom the canal offers great facilities for transit.

Under Rennie and Telford, canal construction was continued, and old methods were improved upon. The Barton aqueduct of Brindley sank into insignificance before the works of these later engineers, whose canals, instead of winding round the hillsides to avoid cuttings, were led through hills and over valleys regardless of obstacles. Besides the completion of English canals, we owe to these two men the construction of the canal from the Forth to the Clyde and the Caledonian Canal, in Scotland; and the two parallel canals in Ireland which connect Dublin with the Atlantic. Thus, in half a century was the country covered with a network of waterways, giving an impulse to manufactures which had hitherto been shut out from foreign markets.

About the end of last century, a great impulse was given to the traffic on the canals by a Mr Baxendale, the agent of Pickford, the well-known carrier. By his efforts, a thorough system of canal communication was established and maintained, and greater punctuality was observed in the arrival and departure of the boats. Express or fly boats also came into use for the more important merchandise and for passenger traffic. On the Bridgewater Canal, they plied with passengers between Manchester and Liverpool; and in the neighbourhood of the larger towns they conveyed the market-women home to the surrounding villages. In 1798, many of the troops for the Irish campaign were conveyed by canal from London to Liverpool. When the railway systems were projected, some of their greatest opponents were the canal Companies, who fancied they saw in the new mode of transit, utter ruin to their own traffic. It was said that the canals would soon become

useless and overgrown with weeds, and it was even proposed to buy up the canal Companies, fill in the water-channels, and lay down the line of rails in their stead. But in spite of all these dark forebodings, and notwithstanding the utility of the new method as compared with the old, the canals still maintain their ground. Their traffic since the advent of the railways has steadily increased; canal shares are usually considered safe stock, and therefore seldom change hands. Both systems of communication have their advantages; and whilst the locomotive is the great economiser of time, there are many articles of commerce, in the shape of building materials and fragile goods, in the carrying of which the canals are more suitable. They remain at the present day a lasting and still useful monument to the English enterprise and perseverance of the last century.

To turn to the present century: M. de Lesseps has been so successful with the Suez Canal, and promises to be with the Panama one, that it is no wonder that he should have many followers; and it is to be noted that the canals proposed now are all on the large scale—canals for ships of large size. They are mostly through narrow necks of land, although one of them is to connect an inland town, Manchester, about thirty miles from salt water, directly with the sea. The Isthmus of Corinth is the site of another; and still another is to run into the great Sahara of Africa and convert it into a great salt-water lake. How long this lake would take to fill up with solid salt is a nice question, which we have not sufficient means of determining, as the other ‘salt lakes’ of the world are all supplied with fresh water, and have only as yet attained to a more or less briny state.

AN IRISH TRAVELLING THEATRE.

MANY people who have heard of a travelling theatre may find perhaps the following peep behind the scenes somewhat interesting.

On a cold, bleak day towards the end of October 1885, I received the following letter:

RESPECTED LADY—I is an actress, and has a travelling theatre. We came to this village two days ago; but the times is bad, and business so slack, I has had to sell most all the theatrical wardrobe; and in consequence we has but little left us we can wear. Respected lady, I writes to ask you to have the harte to help me and my company. Any evening dresses, especial *ballett* dresses, no matter how old, and any artificial flowers, will be thankful received by one who art and health is alike forsakin. Respected lady, I has a large family to provide for, and any old stockings and shoes I pray you to bestow, lady. My daughter is waiting for an answer. We has a benefit for her to-night. Any clothes, lady, looks well on the stage. Reserved seats fourpince, and pit twopince.—Yours respectful to command,

MADOLINE EMERSON,
or MARY FLANAGAN.

I sent for the bearer of the letter, who had, as intimated, waited for my reply. A little girl of about eight years old appeared, and bowed to me very gravely. She was thinly and poorly clad, and looked miserably cold and wretched. Her little feet were without stockings, and red from exposure; they peeped through her broken shoes.

When I asked her would she like some food while she waited, her poor pinched little face brightened as she eagerly said: ‘Yes, lady, if you please. I have had no breakfast, and I am so hungry.’ So, while she partook of the meal she so much needed, I collected what clothes I could, and gave them to her, promising to have some more on the morrow, when I desired her to call again. She did so, bringing with her a letter full of expressions of

gratitude from her mother for the help I had given. It was on this occasion I heard from little Mary the following history of a travelling theatre.

‘We came to this village two days ago. Our theatre is erected in the street, and we call ourselves the Emerson Company. That’s my mother’s name; and it sounds grander-like than my father’s, which is Flanagan. There are six of us alive; but my eldest sister is married these two years, and has a theatre of her own. We mostly marry into the profession, for we find it more useful,’ she added. ‘My big sister at home is fourteen, and we buried two. Next to her, then I come, and I am eight; and my only brother, who comes next to me, is six. No more of us act, because Maggie must mind the baby while mother is acting. My sister dances and sings beautifully; and as for an Irish jig, you never saw the like of her, she’s that good. But she gets frightfully tired, for she has heart disease; and the doctor says as how she may die any minute. I can sing too,’ she continued proudly; ‘and I could dance on the “tight wire” too; but I fell off it two years ago, because I forgot to rub my feet in a white powder we have to use before going on; and since then, I am afraid. But my little brother isn’t, and he can turn a summerset on the wire and juggle grand. He can throw the knives as high as that’—indicating with her hands a distance of three or four feet—‘and can bring the sharp points of the blades on to the palms of his hand without so much as giving them a scratch.’

‘How can he do that, if the knives are so sharp?’

‘Well, you see, lady, father has a big jar of stuff like brown oil—I don’t know its real name—and my brother rubs his hands all over with some of it—very little does; then the knives cannot cut him. It will only come off again by washing his hands in mostly boiling water.’

‘How many are there in your company?’

‘We have only three at present,’ she replied, ‘besides the family. When we want more, my married sister lends us one or two out of her troupe; but of course we pay them. Those we have now act very fair: one gets five shillings a night; and the other two get three shillings and half a crown. If we have a good take at the door, father will give them an extra shilling apiece all round; but some nights they get all we make, and we get none. We only took one pound between these two nights. Business is slack; but maybe we’ll make more soon, when the people in the country hear of us;

for we are a most respectable company,' she added proudly. 'In the last village we were in, we "took" a lot because we had the wonderful speaking pony "Jack." But another company as had a travelling theatre too, came while we were there; and as they were poorer than we were, father, who is real good to any one in the profession, lent them the pony.'

'And what could this wonderful pony do?'

'He could most speak, lady, he was that clever. At Pound's Place—that's where we were afore we came here—we lodged with a grocer in the village. He had a little girl as used to steal sweets out of the bottle from behind the counter in the shop; and the pony found it out, and told on her.'

'How did he do that? Tell me some of this clever animal's tricks.'

'Well, lady, you see, this night father and Jack came on the platform as usual. First, father says: "Now, Jack, who is the biggest rogue in the theatre?" The pony walked round and looked at every one, and then came back and stood before father and nodded his head twice, which meant, "*You* are." But that's only a part of the play, lady; father isn't really a rogue—he's real good. Then father says again: "I wonder, Jack, could you discover who likes a good pinch of snuff?" Jack looked about, and walked a few steps and then stopped before the old woman who sold apples round the corner. 'Twas quite true,' continued the child, 'for she used to buy it where we lodged.—After this, father said: "Now, Jack, as you are so clever, tell the company which of all the little girls present likes sweets, and is in the habit of stealing them?"—and if Jack didn't find out Mollie—that's the little girl as I told you of, lady—and he nodded and nodded his head ever so often, to show he was quite sure it was Mollie! She was very angry, and began to cry, and told Jack as how she didn't steal them. But he knew it was a lie,' added Mary, 'for he would not go away, though father called him. And Mollie she was that mad, she would never again come inside the theatre, she said, because the pony *told* lies of her before every one!

'We have different plays each night, and have beautiful "cuts." Some nights, when the reserved seats are mostly empty, we have only singing and dancing. My sister does a lot of steps then; and when she comes off the stage she is well-nigh dead, she is so hot and tired. Mother is tired every day; for she coughs nearly all night. We are mostly all tired,' the child continued, 'for 'tis twelve o'clock, and often one, before we get to bed any

night. Then there is a rehearsal every day at twelve o'clock. Mother never gets up till 'tis time to go to it.—Our tent was partly blown down last night, lady, for it blew very hard, and it was much damaged. Every strip of canvas costs six shillings, and it takes a great many to make a tent. Mother and the company are mending it now, while I am here.'

'How long will you remain in our village?'

'Maybe a week longer, or maybe two,' answered the child; 'it all depends on the "take" we have. We were six weeks in Pound's Place; but we've only made enough these two nights here to pay the company, and had nothing for ourselves. We are often hungry, Jim and me.'

'Do you like being an actress, and wearing all those bright dresses, and singing for people who applaud and praise you?'

'O no, lady; I hate the life,' she replied; 'and the audience are cross often, if they don't like the piece and what we do; and then I get frightened. Then father sings a comic song, and they all mostly like that.'

'How do you manage to take the tent, its fittings, and your wardrobe about from place to place?'

'We have a big wagon as holds everything, and the horse and the donkey they draw it. Then father hires a car for us, and another for the company, and we travel from village to village that way. We go to the towns in winter. Our theatre is well known; and in some places we make six pounds, and maybe seven or eight, in one night. Other times we might only take—as we've done here—ten shillings. We never go in debt,' she added. 'Mother sells our wardrobe when we are very poor, and then she asks kind ladies to help us by giving us their old clothes. Anything does for the stage so long as it's bright. Once mother got a dress from a lady all over silver stars, and she wore it when she is the Queen. I doesn't mean she is a real queen, but one in the play. But that's worn out now,' she added sadly.

'I must be going now,' Mary said, getting up; 'and I'm very thankful entirely, lady. Maybe you would send the servants to-morrow night to the theatre, for Jim is having his benefit. We don't have any real ladies come, or I'd be real glad to see you,' she concluded ingenuously.

Accordingly, I sent the servants; and from them I heard that the theatre was the most wretched place imaginable. A small tent, in many places broken

and saturated with rain, which had been falling heavily, was pitched in the principal street in the village. A few forms served as reserved seats; whilst those who could not afford this luxury, stood in groups behind. The stage was raised some three or four feet from the ground by means of some barrels, on which long planks of wood were arranged in rows to form a platform. A few candles placed along the edge of it served for footlights; whilst large gaudy ‘cuts,’ representing some specially attractive character in the several plays acted, formed the scenery, as Mary had stated; and on the occasion in question, when singing and dancing were the only entertainments provided, the audience were asked if they wished to come upon the stage and dance an Irish jig or horn-pipe. One man accepted the invitation, and danced both so well and with such a will, amusing the people so effectually, that fully half an hour’s respite was enjoyed by the tired, weary company of the travelling theatre.

INDIAN SERVANTS.

A SOMEWHAT widespread opinion prevails in this country that our Anglo-Indian friends, with their handsome rupee-reckoned salaries, are in the habit of living more than comfortably, if not luxuriously, in the far East. But, in reality, whatever may have been the case formerly, in what were called 'the good old times,' this is not so nowadays; and we should remember that what in England may justly be considered to be a luxury, in a tropical climate like India often becomes a necessity. Our countrymen now—unlike their predecessors, who lived like princes, spent their money freely, and made India their home—wisely adopt the opposite course, and look forward to the time when they may retire on a pension, and pass the remainder of their days in old England.

Perhaps the chief cause which has given rise to the erroneous impression above referred to is the number of native servants which the young Anglo-Indian usually entertains on first taking up his appointment in the Civil Service, the military profession, or other line of business, as the case may be. His mother and sisters are astonished to learn by the first letter received from Jack or Harry—fresh from school, and perhaps hardly out of his teens—that already he has enlisted into his service no fewer than seven or eight attendants; and not comprehending the rights of the case, are apt to moralise on youthful extravagance. This, however, is a mistake on their part, which we will endeavour to explain, at the same time offering a few remarks, for the benefit of our countrymen daily leaving our shores for India, on native servants in general, their duties, peculiarities, and the best way of treating them to meet with success. But before taking them individually, it is with regret we feel compelled to allude to a practice not unfrequently indulged in by the young and thoughtless, of constantly using native terms of abuse to their attendants for the most trivial faults. This is a habit much to be deprecated. The natives of India are extraordinary judges of character, and quickly lose all respect for a master who demeans himself in this manner; and no native servant of any worth will permit himself to be cuffed and knocked about, and, rather than submit to such treatment, will give up his place immediately. The submissive air and humble gait of the natives of

India should alone be sufficient to disarm a European, and prevent him from ever lifting his hand against one of them, even when provoked to the uttermost by some gross act of carelessness or stupidity. A little patience and kindness, coupled with tact and firmness, will generally produce the desired effect, and is much to be preferred to harshness and constant scolding.

The young Anglo-Indian, on reaching his destination at, we will suppose, some up-country station in the North-western Provinces of Bengal or the Punjab, will, generally speaking, require the following servants: a *bearer* or personal attendant; *khitmutghar* or table attendant; *bheestie* or water-carrier; *dhobie* or washerman; *mehter* or sweeper; *syce* or groom; and a grass-cutter to provide fodder for his pony; and throughout the hot-weather months, two additional *coolies* will be necessary to keep the punkah moving throughout the exhausting nights of the tropics.

On first landing from the steamer at the end of the voyage, the young Englishman is sure to be met by numerous applicants for service. The door of his hotel will be thronged by eager candidates for situations; but unless under exceptional circumstances, such as a fellow-countryman travelling homewards, and anxious to obtain a place for a really good servant, he will act wisely to defer making a selection until he has reached his journey's end, when, probably, he will have more time to look around and make his selection.

The first and most important servant to procure is a *bearer*, and it is by no means an easy post to fill up satisfactorily. He should be a Hindu of not too high a caste; nor, on the other hand, of the opposite extreme, a very low caste. The latter is almost certain to prove a failure. There is much to be learned from the personal appearance and style of dress of native servants. Certificates to character should be carefully examined and received with caution; for not unfrequently these documents are forgeries, or borrowed for the occasion; sometimes copies from some genuine certificate supplied to another individual. The applicant for a place should also be questioned on the why and the wherefore of his quitting his last situation. As to caste, perhaps the *kahar* is the best for a *bearer*. Taken generally, the *kahars* are an industrious, quiet race of beings. One of their chief occupations is carrying palanquins; but the opening of railways throughout India has in a great measure done away with this mode of travelling. It may be mentioned

that the title *kahar* many years ago was also the distinctive appellation of a Hindu slave.

As head-servant of the house, the *bearer* should always be well dressed, more especially so as one of his chief duties is to receive visitors at the door. He should never appear without wearing a turban, nor ever enter the house with shoes on his feet. These two latter remarks apply to every class of servant. Nor should a plea of forgetfulness for neglect of the same be ever accepted. The *bearer* is responsible for his master's clothes; he has charge of the keys. He should be the first astir in the morning, and call the 'sahib' at the proper hour to dress for parade, the early walk, or ride. He dusts and arranges the different rooms while his master is out; and on the latter's return has the bath in readiness. With the exception of an hour or two about mid-day, when the *bearer* disappears for his dinner, he remains in the veranda or within call. He keeps account of small household expenditures, again attends his lord on the latter retiring to rest, when the *bearer* makes his final salaam or obeisance, and takes his departure.

The next in importance among Indian domestic servants is the *khitmutghar* or table-attendant. It need hardly be mentioned that he is invariably of the Mohammedan religion; and great care is necessary in choosing this particular servant, for among their ranks are many low, dissipated characters. A single glance at one of these latter will generally suffice to make one aware of the fact. Old graybeards, though of course less active than younger followers of the Prophet, yet often prove to be better servants in the long-run. When questioned, these gentry almost invariably deny all knowledge of the English language; but, generally speaking, the Bengal *khitmutghar*, as he stands with folded arms and imperturbable countenance at the dinner-table, readily follows and fully comprehends the topics of conversation carried on by his English masters.

The duties of the *khitmutghar* commence at daylight, when he puts in an appearance bearing the morning cup of tea. Unless otherwise ordered, he is only expected to be present, properly dressed, at each meal. One of his most important duties is to be able to cook fairly well when called upon to do so, more especially when his master may move into camp either on the march or on a shooting expedition. Then he is expected to show his powers in the culinary art; and, generally speaking, Mohammedan cooks acquit themselves admirably in this respect. They are especially clever at making

omelettes, soufflets, and such-like. It may be here mentioned, however, by way of warning to the uninitiated in such matters, that the native method of preparing a meal is not always too nice to our ideas, so that it is well to avoid visiting the cooking-tent immediately before dinner, or not improbably you will there see something or other going on not calculated to give one an appetite.

Next in our list comes the *bheestie* or water-carrier, also of the Mohammedan religion; but altogether a less troublesome mortal to deal with. Generally speaking, the Bengal *bheestie* is a good, willing, hard-working servant, seldom giving trouble or requiring reproof. His chief duties are to supply the house and stables with fresh water from the best well in the neighbourhood. It is the special duty of the *bheestie* to keep the *chatties* or earthen jars of the bathroom filled with water. Where a garden is kept up—and in hot climates there is nothing so refreshing to the eye as a few flowers and bright-green shrubs around the house—it is the duty of the *bheestie* to assist the native gardener in watering the plants. He also, morning and evening, sprinkles with water the flooring of the verandas, footpaths, and dusty roads in the vicinity of his master's abode. This has the effect of laying the dust and cooling the air—no slight boon to exhausted Europeans during the terrible months of April and May, just before the first rainfall.

The *dhobie* or washerman is another important individual in the Anglo-Indian establishment. The great majority of *dhobies* are Hindus; but in Eastern Bengal, Mohammedan *dhobies* are often to be met with. Though given to assuming airs and importance, the *dhobie* is of low caste, generally speaking; a mild inoffensive being, plying his trade industriously, and giving little trouble to his master. There is a proverbial saying that obtains among the Hindus which pronounces a *dhobie* as untrustworthy; but in reality he is no worse than his brethren in this respect. The *dhobie* is one of the first to bestir himself in the early morning, and accompanied by a small *bail* or bullock, carrying his bundle of clothes, he may be seen making his way in the direction of some tank or distant pool on the river-bank. On reaching the scene of operations, he strips himself of superfluous clothing, girds up his loins, and proceeds to business. Soon the air resounds with the heavy thwacks of some article of raiment, which, twisted into a small compass, the *dhobie* again and again whirls round his head, and brings

down upon a flat piece of wood or stone placed on the margin of the water. Each blow is accompanied by a grunt from the operator, as if to give an additional impetus to the stroke. This somewhat rough treatment is liable to wear out fine linen all too soon, and to make buttons fly; but considering that the *dhobie* has no mangle to assist him, nor any of the ordinary appliances of a laundry, and, generally speaking, only a small smoky hovel—probably filled to overflowing with his wife and numerous children—wherein to complete his work, it is astonishing how well he acquits himself of his task; the well-starched, snow-white shirt-fronts bearing witness to his skill and painstaking. Unless articles of clothing are plainly marked, the *dhobie* has a tiresome habit of sewing coloured pieces of cotton into the corners of every shirt and handkerchief, to distinguish them from others, which practice has anything but a beautifying effect. The *dhobie* considers himself so far independent that he need only appear at stated times, to receive or make over his master's clothes from the hands of the *bearer*. He will never take service as an indoor servant in the house of a European.

The duties of the *mehter*, sweeper or 'knight of the broom,' are so commonplace as to require only a brief notice. He is always of low caste; and though often addressed as 'jemadar' by the other servants, he is always looked down upon, more especially for his habit of eating or drinking anything left from the table of his master. It is his special duty to take charge of and feed his master's dogs. He supplies them with food at a fixed rate, takes them in the early morning for a bathe in the nearest tank, and towards sunset, produces for inspection, in separate iron dishes, the food which he has provided for each one of his charges.

It is amusing to observe how well-bred English dogs despise and turn up their noses at their native attendant, permitting the latter to lead them about and wash them when necessary without a growl of disapproval, but at the same time clearly showing by their outward bearing that no familiarities will be permitted.

Next we come to the *syce* or native groom; and in a stable where a valuable Arab horse has to be cared for, he is a most important personage. A really good, trustworthy *syce* is nowadays seldom to be met with. There are Mohammedan *syces* throughout Northern India; but the great majority are Hindus of low caste. The duties of the *syce* are, to groom and feed the horse he is put in charge of—a separate *syce* is necessary for each one of the

horses comprising a stable—to be ready to accompany his master to the parade-ground, the band-stand, or for wherever he may be bound; and to keep the latter in sight and follow him any distance, no matter at what pace the sahib may choose to ride. It is astonishing what powers of endurance these native grooms display in this respect; for however far the distance or quick the gallop, he is seldom left far behind, and nearly always makes his appearance soon after his master draws rein.

A Bengal syce worthy of the name can hardly in any country in the world be surpassed at his work. He is a most excellent groom; and by means of hand-rubbing—which he often practises for hours together—he brings out the muscles and sinews of a horse till they are as tough and hard as iron. It is a good custom to inspect daily the allowance of corn or grain provided by the syce for his charge, as not unfrequently dishonest grooms steal a portion of it and grind it for their own food.

The ‘grass-cutter,’ the last in our list, is a humble individual, who, as his title tells us, supplies grass for the horse to which he is attached. Hay is seldom seen in India; but horses thrive well on a particular kind of soft green grass, which the grass-cutter cuts, or rather digs up with a small iron instrument called a *koorpah*. It is well, every now and again, to examine the quality and quantity of the grass supplied for each horse, or else lazy individuals will likely enough bring in coarse hard stuff quite unfit for the purpose. In large stations, a grass-cutter who performs his work properly has often to walk many miles before reaching a spot where soft tender grass is procurable. The grass-cutter is under the immediate orders of the syce, and usually receives four rupees a month as pay for his services.

In conclusion, it may be mentioned, that one of the most important rules in the young Englishman’s household should be that each native servant regularly receives his pay on a certain date in each month. Without this being steadily acted up to, matters never work smoothly in an establishment, but will cause constant bickerings. Whereas, when paid regularly, and treated with kindness and forbearance, these poor people speedily become attached to their master, and exert themselves to meet with his approval.

J. H. B.

THE MONTH: SCIENCE AND ARTS.

ACCORDING to *Nature*, the much-dreaded scourge of the vine, the Phylloxera, has made its appearance in the vineyards of the Cape Colony. Some years ago, the most stringent regulations were made to prevent, if possible, the importation of these unwelcome guests. The Cape government even refused to allow consignments of beech-trees from England and tree-ferns from New Zealand to be landed in the colony, and fixed a very heavy penalty as a punishment for any infringement of the law. But by some mysterious agency, two or three of the vineyards are swarming with the Phylloxera. The most approved insecticides, carbon disulphide, &c., have been telegraphed for, for they are not at hand in the colony, and in the meantime the affected vines are being uprooted and burnt.

A curious instance of tenacity of vitality in low forms of life has been discovered by Professor Leidy. Upon examining a block of ice which formed part of a large quantity stored at Moorestown, N.J., and had been so stored for more than twelve months, he found it riddled with air-bubbles and drops of water. Upon melting a portion of the block, a number of worms made their appearance. They died almost immediately when liberated from their frozen prison. The worms cannot be identified with any known species, and Professor Leidy believes them to be of a form as yet undescribed.

It is satisfactory to note that the Emperor of Brazil has given orders for a photographic astronomical apparatus like that employed so successfully by MM. Henry of Paris, in order that Brazil may do its share in the proposed photographic survey of the heavens initiated by the French astronomers named.

Lloyd's agent at Athens has recently reported some information as to the progress of the canal which is to cut the Isthmus of Corinth. Out of a total of thirteen million cubic yards of earth which must be dug out before the canal is completed, nearly three millions have been removed. The canal is to have a surface width of twenty-four yards, except at the entrances, which

will be widened to between fifty and sixty yards. One thousand men are at present employed upon the works, which, at the present rate of progress, should be completed in five years.

The Austrian government offer a prize of one thousand ducats (nearly five hundred pounds) for the discovery of a system of working coal in fiery mines without shot-firing. The method must not be more expensive than that of ordinary blasting. It must not be capable of igniting fire-damp or coal-dust, and it must not leave any injurious products behind it. These are the chief conditions.

An improved method of etching metallic surfaces has been invented by Mr A. Piper of Wolverhampton. The metal surface is first of all coated with gold, silver, nickle, brass, or any other metal desired, in the ordinary electro-plating bath. The design is then drawn upon it in some resinous or other acid-resisting medium, and the metal is immersed in an acid, which eats away the coating, and at the same time produces a dead or frosted appearance upon the exposed metal beneath. The resinous drawing is now removed by any suitable medium which will dissolve it, leaving the design in relief upon a frosted ground. If desired, the operation can be reversed by leaving a groundwork of plated metal, while the design is bitten out by the acid.

A new stationary buffer-stop for railway stations and sidings was recently described in a paper read before the Institution of Mechanical Engineers by Mr A. Langley of Derby. This buffer-stop consists of two hydraulic cylinders fitted with pistons. The piston rods carry in front buffer-heads to meet those on the locomotive. There are also projecting rods behind the cylinders connected by chains with counterweights, to return the pistons after pressure to their former position. The pistons have a stroke of four feet; and it is calculated that this amount of depression would effectually stop a train without jerk or damage even if it were moving at the rate of eight miles an hour.

Gases inclosed in iron cylinders under enormous pressure are now used in various branches of science and art, and are supplied commercially by many firms in various countries. The gases most commonly used are hydrogen and oxygen—for the lime-light—carbon dioxide, and nitrous oxide—this last being much employed as an anæsthetic by dentists.

Hitherto, there has been much difficulty in controlling the outrush of gas from these cylinders, for the internal pressure often amounts to six hundred pounds on the square inch. By the aid of a new regulator, invented by Messrs Oakley and Beard of London, this difficulty is at once obviated. The regulator consists of a small india-rubber bellows inclosed in a brass box, which screws upon the nozzle of the gas cylinder. By an ingenious device, as the bellows top rises with the pressure of the gas, a screw valve descends upon the opening in the cylinder. In this way the user of the gas can regulate the outflow to his requirements. We understand that it is in contemplation to adapt the same principle to ordinary gas consumption in houses, so that the supply may always be adjusted to the number of burners in actual use.

Habitual drinkers of aërated beverages were some time ago startled by the report that the original source of the water used in the manufacture did not much trouble the attention of the vendors, and that micro organisms in fabulous numbers might find their way to the consumers of these apparently innocent fluids. According to Dr T. Leone's researches, aërated waters are peculiarly safe from such contamination. Taking a typically pure potable water, he tried how many micro organisms could be developed in it in a given time. In five days the water contained immense numbers of organisms. But when charged with carbonic dioxide, as all aërated waters must be to give them their effervescent quality, the number of living creatures was at once diminished. Water so charged contained at the end of fifteen days only a mere trifle of the original organisms. Dr Leone therefore concludes that the longer aërated waters are kept, the less chance is there of bacterial contamination.

The greatest living authority on bacteria, M. Pasteur, has by recent experiments proved that water containing only two per cent. of concentrated sulphuric acid possesses the property of destroying these organisms. He recommends that this acidulated water should be used as a disinfectant for floors of stables, mangers, courtyards, cattle-sheds, &c. The compound has certainly the merit of extreme cheapness, for about twelve gallons could be prepared at a cost of twopence. We may mention that M. Pasteur's inoculations for hydrophobia have met with unlooked-for success. He recently told the Paris Academy of Sciences that out of three hundred and twenty-five cases of inoculation for this terrible disease, only one had proved a failure, and that one he attributed to delay. It is suggested that an

international hospital should be established for the reception of patients from every country.

The all-seeing microscope has very often played an important part as an accusing witness, more especially in the identification of blood-stains. Recently in Illinois the same detective agent was instrumental in hanging a murderer; but the method of conviction was novel. Here is the case: A. had been found murdered while sleeping on a pile of sawdust in a certain icehouse, which we will call No. 1. B. was suspected of the crime because particles of sawdust were found on his clothing and on his boots. He accounted for this by pleading that he had been sleeping in another icehouse (No. 2) which was far away; and declared that he had not been near the No. 1 house. It was proved that icehouse No. 1 contained *pine* sawdust, and house No. 2 hardwood sawdust only. The microscope showed that the clothes and boots had attached to them particles of the former only. The man was convicted and executed.

Professor Vogel has lately brought forward the curious fact that the generation of alkaloids in plants is dependent upon sunlight. The hemlock plant which yields coniine in Southern Europe contains none in Scotland. Again, the tropical cinchonas, from which quinine is obtained, will yield very little of that valuable product if cultivated in our weakly lighted hothouses. Professor Vogel has examined many specimens of the plant from various conservatories, and has been quite unable to obtain the characteristic reaction of quinine, although the method of testing is a delicate one, and sensitive to minute quantities of the alkaloid. It is curious to observe that although sunlight seems so necessary to the formation of quinine in the living plant, it acts most injuriously upon the alkaloid in the stripped bark. In the latter case, the quinine is decomposed by it, and assumes the form of a dark-coloured resin. Because of this, in the manufacture of quinine, the bark is always dried in the dark.

The recent severe weather must have led many a half-frozen traveller to wonder if our railway and tramway Companies will ever hit upon some method of heating public conveyances. With a steam-engine as a necessary adjunct, it would seem to the disinterested inquirer that a method of warming by pipes fed from the 'exhaust' would be a comparatively easy way of managing the business, and would at the same time save much labour in doing away with the filling and distribution of inefficient foot-

warmers. In Chicago, a new method of heating tramcars is being tried, and it promises well. The apparatus, which is placed under the floor of the car, consists of a brass cylinder filled with coal-oil, which, under pressure of a strong spring, is forced into a small super-heater, where it becomes vaporised. This oil-gas is ignited in a fire-clay combustion chamber, and although there is no flame, the fire-clay is brought to a white heat. The outer air passing over this hot box becomes well warmed, and a constant stream of fresh, warm air is assured to the passengers. The only visible evidence of the stove is a grating in the floor of the car through which the hot air rises. When will our tramway Companies consent to a small reduction in their high dividends, to afford their patrons similar comfort?

The use of wood pavements in Sydney has been very strongly condemned by a Committee appointed by the Legislative Assembly of New South Wales to inquire into the subject. It had been alleged that these wood pavements exerted a deleterious influence on the health of persons living in their proximity, and the conclusions arrived at by the Committee would seem to justify these allegations. Analysis showed that the blocks in actual use had absorbed a vast amount of organic matter, even though they had only been laid down a comparatively short time. It was evident, too, that complete impregnation of the wood was only a matter of time. In the words of the Report: 'So far as the careful researches of your Board go, the porous, absorbent, and destructible nature of wood must, in their opinion, be declared to be irremediable by any process at present known; nor, were any such process discovered, would it be effectual unless it were supplemented by another which should prevent fraying of the fibre.' It should be noted that this strong condemnation is applied to the *hard* wood-blocks used for the purpose of paving in Sydney, and not to the soft wood used here at home. These latter are so thoroughly impregnated with tar, that it is difficult to imagine that room could be found for anything else, organic or otherwise.

The cable tramway which is situated on the historic hill at Highgate, London, has worked without hindrance during the recent frost and snow. This is due to the fact that the working parts are underground. But of late a new use has been made of the system. Heavy vehicles even with six horses attached could not be moved up the hill during the recent frosts. Many of

them were therefore fastened to the tramcars, and were pulled up the steep incline—one in eleven—at the rate of six miles an hour.

In his recent lecture at the Royal Institution, Mr A. A. Common, the treasurer of the Astronomical Society, pointed out that the old method of eye-observation in telescopic work would probably in the near future give way to automatic records on sensitive dry plates by means of photography. He also pointed out what extreme variations existed in the amount of light emitted by different celestial objects, contrasting with the blinding glare of the sun the small quantity of light received from a faint star. The latter he described as being equivalent to the twenty-thousandth part of the light given by a standard candle seen from the distance of a quarter of a mile. It is not a matter for wonder, therefore, that the most sensitive dry plate which would yield a photograph in daylight in the smallest fraction of a second, should require an exposure of two hours, or thereabouts, when used for recording the existence of one of these distant orbs.

Signor Ferrari, after making observations on between six and seven hundred thunderstorms which occurred in Italy a few years back, has noted that every thunderstorm is connected with a barometric, hygrometric, and thermic depression. A German scientist who has interested himself in the same line of inquiry, states that the danger of a building being struck by lightning has increased in his country during the past half-century from three to five fold. He attributes this increase of danger to impurities carried into the atmosphere from factory chimneys, the number of which is constantly increasing.

A new electric alarm-bell for use in places where highways and railways cross one another has been invented in the United States. On approaching such a crossing, the wheels of the train depress a heavy trigger placed by the side of the rails. This trigger sets in motion a flywheel sufficiently powerful to turn the armature of a small magneto-machine. The current thus generated rings a bell at the cross-road, so that wayfarers have an audible reminder of the near approach of a train. Of course the same result might be brought about with an electric battery. But the magneto-machine has the advantage of requiring no attention, and of not being affected in any way by changes of temperature. Its bearings can be provided with oil-cups, so that it will act for months together without supervision.

Professor Ewart lately read a paper before the Royal Physical Society of Edinburgh, in which he stated that from examinations of specimens of 'whitebait' sent into the London market during the past year, he had come to the conclusion that the much esteemed little fish consisted of sprats and herrings, about sixty per cent. of the former to forty per cent. of the latter. The origin of so-called 'whitebait' has always been such a disputed point that the Professor's remarks are particularly interesting. He also pointed out that in Canada, sprats are extensively tinned as sardines. As we some time ago pointed out in these pages, a similar trade has been for a long time carried on at more than one place on the south coast of England. We may mention that the authorities of the South Kensington Aquarium are about to introduce herrings into the tanks under their control, in spite of the fact that all such attempts, in inland places at least, have hitherto failed. We trust that their endeavours will be crowned with success.

Messrs Fairbairn and Wells, Manchester, have lately much improved their screw forging machine. By this method of making screws, it is claimed that much greater tenacity, ductility, and durability are obtained in the finished product; for the fibres of iron, instead of being cut through, are pressed and bent round to the shape of the thread. In short, the machine rolls out the screws, instead of cutting them out. We have lately seen photographs of some of these screws which have been partially eaten away with acid, for the purpose of showing the fibrous nature of the metal. It is curious to note how the fibre is bent in and out as it follows the direction of the thread on the screw. This method of manufacture is said to present advantages apart from better quality. The screws can be more quickly produced at a less cost, and there is a great saving of material, for nothing is cut away to waste.

The results of a curious but very important test as to the accuracy with which chemists, druggists, and others make up prescriptions committed to their care, has recently been presented to one of the London vestries. Fifty prescriptions were sent out to ordinary druggists, to co-operative stores, to 'doctors' shops,' and to certain traders styling themselves drug Companies. The mixtures made were afterwards analysed, to find out how nearly they agreed with the prescriptions they represented; but in order to give a liberal margin for error, it was resolved not to put a black mark against any one, if the chief constituent were within ten per cent. of the right amount. Notwithstanding this margin, no fewer than seventeen out of the fifty

mixtures were incorrectly dispensed. In one case the principal drug was less by eighty-five per cent. than the amount ordered, while in another it was fifty-seven per cent. in excess. The chemists and druggists came out best in this strange competition, as only six per cent. of their prescriptions had to be called in question. Next came the co-operative stores with twenty per cent. of error; then the 'doctors' shops' with fifty per cent.; and lastly the drug Companies, who are credited, or rather discredited, with seventy-five per cent. of errors.

OCCASIONAL NOTES.

THE ROOSEN METHOD OF PRESERVING FISH.

WE have already noticed the method recommended by Professor Cossar Ewart of preserving fresh fish with boracic acid and salt. Mr Roosen of Hamburg has patented another method, which was lately tested in Edinburgh. The process consists in the salmon being placed in an air-tight compartment among a solution of boracic acid, salt, and water, and a heavy pressure being applied, the solution penetrates and thoroughly disinfects the fish, which are prevented from decaying, and retain all their strength and nourishment. On the 15th of February, a steel barrel, made for the purpose, and capable of holding about three hundred pounds of fish, was filled with salt water containing about fifty per cent. of boracic acid, and into this compound five splendid salmon, fresh from the Tay, were placed. The air having been entirely withdrawn, the barrel was hermetically sealed, a pressure of six atmospheres, or ninety pounds to the square inch, being applied. After standing for seventeen days, the barrel was opened on the 4th of March, and all the salmon were found in as fresh and healthy a condition as when they were first placed among the solution. The flesh of the fish was of a beautiful colour, and could not be distinguished from that of a fresh salmon placed alongside of it, while the blood began to flow freely immediately on the salmon being cut up. The salmon was served up at a luncheon, on the following day, in different forms of cooking, and the general opinion was that the new method of preservation was upon the whole successful. The fish was of good flavour and colour; it could be separated in nice flakes, and the curd was well preserved.

A NEW BOILER-FEEDER.

The problem of feeding boilers has yet to be solved, no method yet introduced having by its intrinsic merits superseded all others. At the present time, injectors or donkey-engines are usually employed—the former being an ingenious apparatus which forces the feed-water into the boiler by

the rush of steam through a narrow orifice; the latter differing in no way from an ordinary pumping-engine, and usually deriving its power from the boiler itself. Both these systems of feeding require constant supervision; and it is to obviate the necessity of continued attention and the risk resulting from carelessness, that boiler-feeders automatic in action have been designed. So far, their application has not been extensive; but a recent improved design, patented as 'Mayhew's Automatic Boiler-feeder,' bids fair to push its way even in these times of depressed trade, when boiler-owners not unnaturally hesitate to incur any outlay, however slight, which a rearrangement of the boiler-feed necessarily entails. The apparatus consists essentially of two vessels—the upper of copper, the lower of cast-iron. An ingenious valve-arrangement connects the former with the boiler, whilst the latter is connected with the supply of feed-water. When the water in the boiler falls below a certain level, the end of the pipe connected with the upper vessel becomes uncovered, and the steam being free to enter it, operates on the valves, thereby admitting a charge of water to the boiler from the copper vessel. A vacuum is formed in the copper vessel, which now recharges itself from the one beneath, ready for another operation. As many as five charges a minute can thus be obtained. Should the apparatus, from any cause, fail to work, and the water fall too low, a fusible plug melts and sounds an alarm whistle. It is satisfactory to note that an ingenious straining arrangement works well for feeding with dirty water—the great difficulty in all apparatus of this class, owing to the valves becoming choked. The feeder may be regarded as safely beyond the mere experimental stage, a large number already being in operation in different works throughout the country, and with results satisfactory in every respect.

SHOT-FIRING IN COAL-MINES.

A correspondent thus writes: 'In your article on "Shot-firing in Coal-mines" you speak of the dangerous operation of tamping or plugging the shot-hole with brick or coal dust rammed hard. It must indeed be a dangerous operation; but cannot the hole be as effectually plugged without any danger at all? It is usual, after charging a rocket, to drive in dry clay upon the top of the fuse, to prevent its blowing through; but a layer of wet plaster of Paris poured in and allowed to set, dries harder than the clay, and obviates all danger from concussion or grit. Cannot the shot-hole be in like manner

plugged? Plaster of Paris (gypsum or sulphate of lime) expands, not shrinks, when combined with water, so that it fills accurately every part of the bore. If the hole were slightly conical, the smaller end outwards, or made with an internal flange, the plaster would offer more resistance than the clay.'

PARTED.

THE silver brooks will miss thee,
The breeze that used to kiss thee,
And ruffle with a soft caress thy curls of sunny hair;
When the early dewdrops glisten
On the roses, they will listen
For thy step upon the garden walk, thy laughter in the air.

The meadows gay with flowers,
The summer's leafy bowers,
Will know thy joyous smile no more; the woodlands stand forlorn;
I hear the soft complaining
Of birds, from mirth refraining.
That greeted with their carols sweet thy waking every morn.

Poor mother! hush thy weeping
Above thy darling sleeping,
Nor fret with aught of earthly grief the stillness where he lies,
Flowers in his little fingers,
Where the rosy flush still lingers,
For the angels are his playmates on the plains of Paradise.

J. I. L.

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